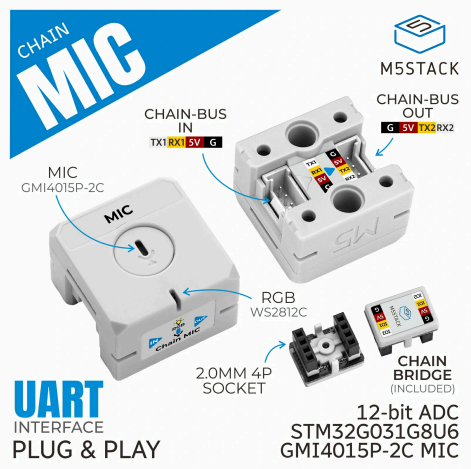
# Chain MIC

SKU:U226
































## Description

Chain MIC is a sound acquisition node in the M5Stack Chain series. It features a built-in GMI4015P-2C microphone, an SGM321YC5/TR operational amplifier, and a programmable RGB LED. It supports **12-bit ADC** sampling of sound signals and threshold triggering, making it suitable for applications such as environmental sound detection, voice-controlled interaction, and device status indication. It integrates an STM32G031G8U6 main controller and uses a UART serial daisy-chain communication protocol. Its two HY2.0-4P expansion interfaces allow more Chain series devices to be connected, enabling richer sound sensing and interactive applications.

## Features

- M5Stack Chain series
- GMI4015P-2C microphone sound acquisition
- Supports **12-bit ADC** sampling of sound signals and threshold triggering
- STM32G031G8U6 main controller
- 1 x RGB LED
- UART serial daisy-chain communication protocol
- 2 x HY2.0-4P expansion interfaces for connecting Chain series devices

## Includes

- 1 x Chain MIC
- 1 x Chain Bridge

## Applications

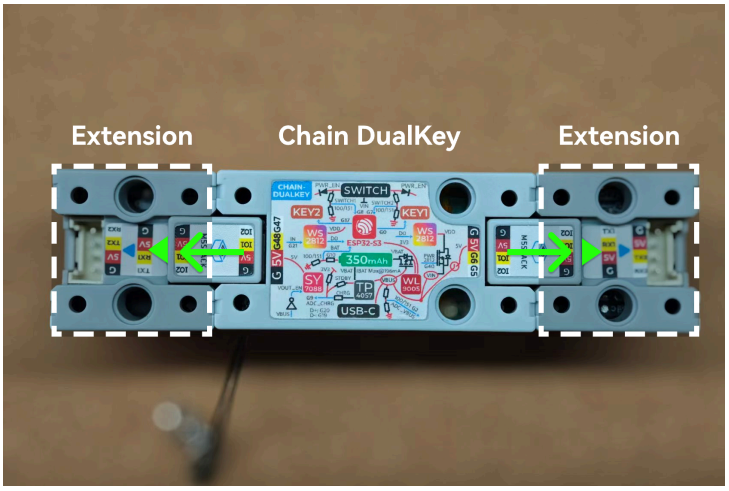
- Environmental sound detection
- Sound threshold triggering
- Voice-controlled interactive devices
- Smart home sound sensing

## Specifications

| Specification | Parameter |
| --- | --- |
| MCU | STM32G031G8U6 |
| Power Input | DC 5V |
| Communication | UART 115200bps@8N1 |
| Interface | 2 x HY2.0-4P |
| Microphone | GMI4015P-2C |
| Operational Amplifier | SGM321YC5/TR |
| RGB LED | 1 x WS2812C-2020 |
| Sound Signal | 12-bit ADC sampled value, threshold trigger status |
| Operating Power Consumption | 5V@35.54mA (MIC active, white LED always on) |
| Product Size | 24.0 x 24.0 x 16.0mm |
| Product Weight | 5.7g |
| Package Size | 138.0 x 93.0 x 17.0mm |
| Gross Weight | 9.0g |

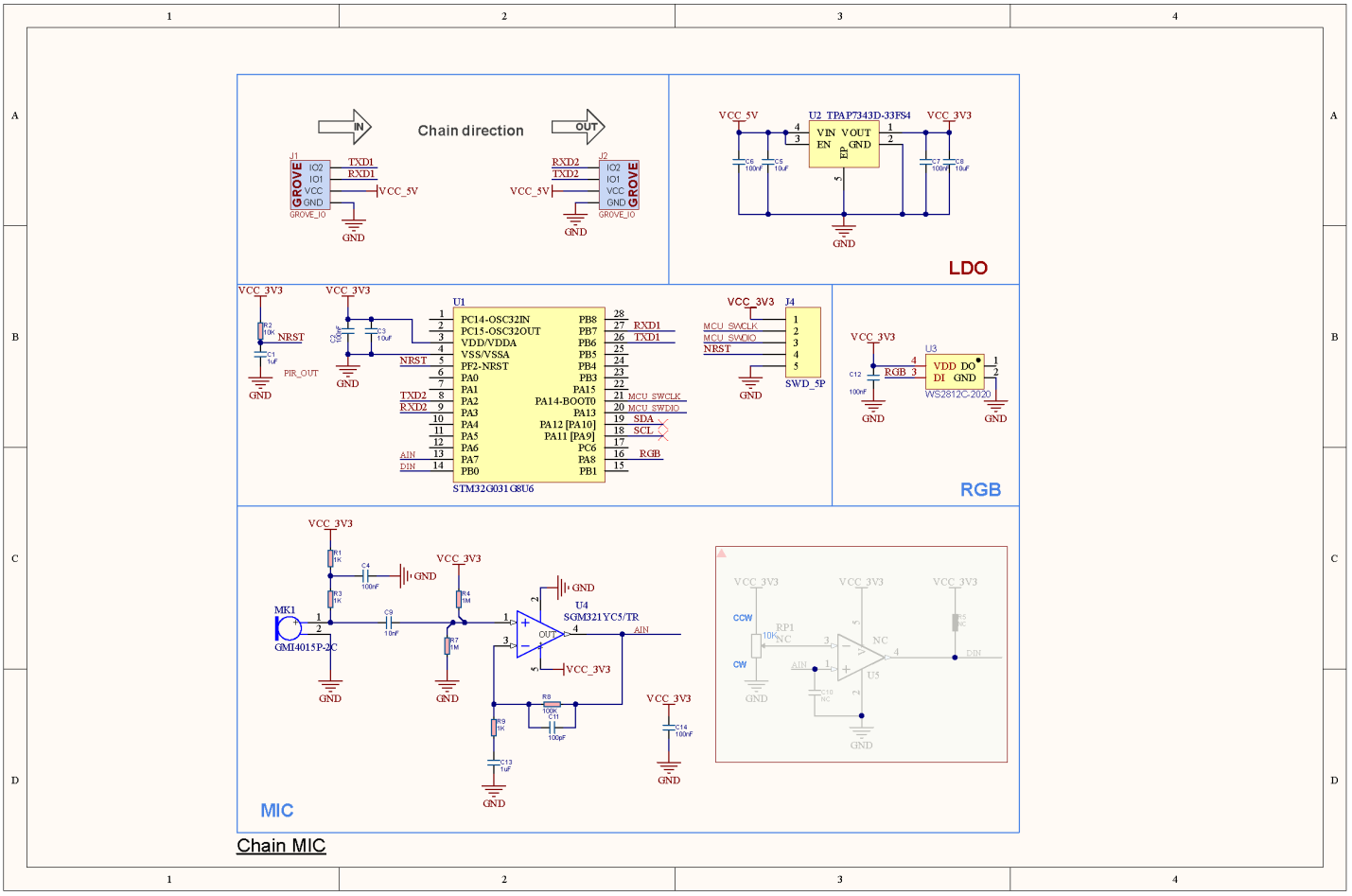
## Learn

Use Chain Bridge connectors to connect the Chain DualKey controller to each Chain series device. Pay attention to the orientation when connecting: the triangular arrows should point outward from the Chain DualKey controller, as shown below:




## Schematics

- Chain MIC Schematics PDF




# PinMap

## RGB LED

| STM32G031 | PA8 |
|---|---|
| WS2812C | RGB |

## MIC

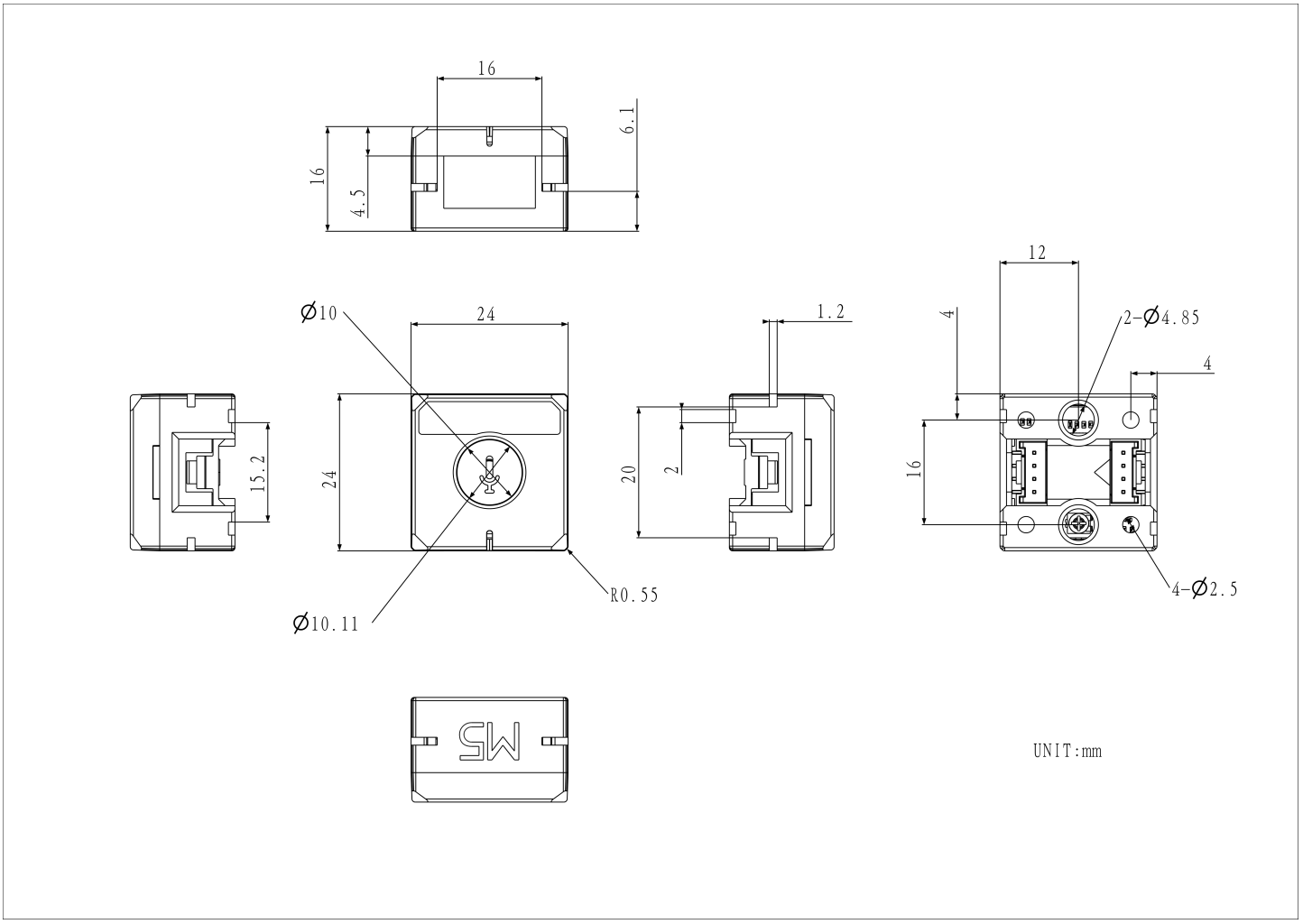
| STM32G031 | PA7 | PB0 |
|---|---|---|
| MIC | AIN | DIN |

## UART

| STM32G031 | PB6 | PB7 | PA2 | PA3 |
|---|---|---|---|---|
| UART1 | TXD1 | RXD1 | | |
| UART2 | | | TXD2 | RXD2 |

# Model Size

- Chain MIC Model Size PDF




# Softwares

## Arduino

- Chain MIC Arduino Quick Start
- Chain MIC Arduino Driver Library

## UiFlow2

- Chain MIC UiFlow2 Documentation

## Internal Firmware

- Chain MIC Internal Firmware

## Protocol

View Protocol

# Video